

torch.aminmax#

<https://pytorch.org/docs/stable/generated/torch.aminmax.html#torch.aminmax>

Description:

Computes the minimum and maximum values of the input tensor. input (Tensor) – The input tensor dim (Optional[int]) – The dimension along which to compute the values. If None, computes the values over the entire input tensor. Default is None. keepdim (bool) – If True, the reduced dimensions will be kept in the output tensor as dimensions with size 1 for broadcasting, otherwise they will be removed, as if calling (torch.squeeze()). Default is False. out (Optional[Tuple[Tensor, Tensor]]) – Optional tensors on which to write the result. Must have the same shape and dtype as the expected output. Default is None.

Function Signature

```
torch.aminmax(input, *, dim=None, keepdim=False, out=None) -> (Tensor min, Tensor max)#
```

Parameters

Keyword Arguments

Returns

Raises

Parameters

Keyword

dim (Optional[int]) – The dimension along which to compute the values. If None, computes the values over the entire input tensor. Default is None.

keepdim (bool) – If True, the reduced dimensions will be kept in the output tensor as dimensions with size 1 for broadcasting, otherwise they will be removed, as if calling (torch.squeeze()). Default is False.

out (Optional[Tuple[Tensor, Tensor]]) – Optional tensors on which to write the result. Must have the same shape and dtype as the expected output. Default is None.

Returns:

A named tuple (min, max) containing the minimum and maximum values.

Code Examples

```
>>> torch.aminmax(torch.tensor([1, -3, 5]))
torch.return_types.aminmax(
  min=tensor(-3),
  max=tensor(5))

>>> # aminmax propagates NaNs
>>> torch.aminmax(torch.tensor([1, -3, 5, torch.nan]))
torch.return_types.aminmax(
  min=tensor(nan),
  max=tensor(nan))

>>> t = torch.arange(10).view(2, 5)
>>> t
tensor([[0, 1, 2, 3, 4],
        [5, 6, 7, 8, 9]])
>>> t.aminmax(dim=0, keepdim=True)
torch.return_types.aminmax(
  min=tensor([[0, 1, 2, 3, 4]]),
  max=tensor([[5, 6, 7, 8, 9]]))
```

Notes & Warnings

NOTE

Note NaN values are propagated to the output if at least one value is NaN.

NOTE

See also `torch.amin()` computes just the minimum value `torch.amax()` computes just the maximum value

Detailed Documentation

Documentation

Computes the minimum and maximum values of the input tensor.

`input (Tensor)` – The input tensor

`dim (Optional[int])` – The dimension along which to compute the values. If `None`, computes the values over the entire input tensor. Default is `None`.

`keepdim (bool)` – If `True`, the reduced dimensions will be kept in the output tensor as dimensions with size 1 for broadcasting, otherwise they will be removed, as if calling `(torch.squeeze())`. Default is `False`.

`out (Optional[Tuple[Tensor, Tensor]])` – Optional tensors on which to write the result. Must have the same shape and dtype as the expected output. Default is `None`.

A named tuple (`min`, `max`) containing the minimum and maximum values.

`RuntimeError` – If any of the dimensions to compute the values over has size 0.

`NaN` values are propagated to the output if at least one value is `NaN`.

`torch.amin()` computes just the minimum value `torch.amax()` computes just the maximum value